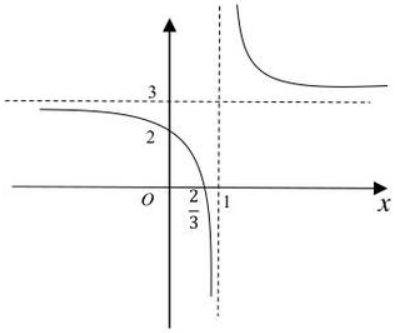
Type MS · Sketching a rational curve with its asymptotes

Answer 1 of 17 · Question 9 part b · 5 marks

2020 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

9(a)(i)	$\frac{2}{p} = 2$ so $p = 1$ *	M1A1cso
(ii)	$\frac{dy}{dx} = \frac{q(x-1)-(qx-2)}{(x-1)^2}$	M1A1
	When $x = 0$ $\frac{dy}{dx} = \frac{-q+2}{1} = -1, \Rightarrow q = 3$	M1A1,A1 (7)
(b)		B1ft B1ft B1 B1ft B1ft (5)
(c)	$x + 2 = \frac{3x-2}{x-1}$	M1
	$x^2 - 2x = 0$	M1
	$x(x-2) = 0$ $x = 2$	dM1A1cao(4) **[16]**

(a)(i)M1	Set $x = 0$ in the curve equation and equate result to 2. Obtain a value for p.
A1cso	Correct value of p obtained from a correct equation.
(ii)M1	Attempt the quotient rule. (formula is given on formula page). Denominator must be $(x-1)^2$. Numerator to be $q(x-1)-(qx-2)$ or $(qx-2)-q(x-1)$ **Must use $p = 1$ now or later.**
A1	Fully correct derivative
ALT	Use product rule: $\frac{dy}{dx} = q(x-1)^{-1} - (qx-2)(x-1)^{-2}$ M1 for attempt with 2 terms similar to above, either term to be correct A1 Both terms correct
M1	Set $x = 0$ in their derivative and equate to -1
A1	Correct equation
A1	$q = 3$

Type MS · Sketching a rational curve with its asymptotes

Answer 1 of 17 · Question 9 part b · 5 marks

2020 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
(b)	**No value for *q***: B0B1B0B1B1 available. **Incorrect *q*:** B1B1B0B1B1 available.	
B1ft	Equations of asymptotes seen or lines parallel to axes passing through $x = 1$, $y = 3$ drawn. $y = 3$ or their q. **Must have a value for *q*.**	
B1ft	Coordinates of crossing points seen explicitly or marked on the sketch. Must have $y = 2$; may have $x = 2 / q$ (value for q not needed)	
B1	Two branches in the correct "quadrants" Must have $q = 3$ for this mark. Asymptotes drawn.	
B1ft	There must be at least one branch of the curve drawn and 2 asymptotes drawn and labelled on the diagram by showing the coords of the points where they cross the axes or with their equations. The curve must not touch (or cross) either asymptote. ft their asymptotes**, inc *y* = *q***	
B1ft	Both crossing points clearly marked on their diagram. ft their crossing points .	
(c)		
M1	Eliminate y between the line and the curve equation. May use q or their value for q	
M1	Obtain a 2 or 3 term quadratic. May use q or their value for q.	
dM1	Solve their equation to obtain 1 or 2 values of x Depends on both M marks above.	
A1cao	$x = 2$ from a correct equation. If $x = 0$ is seen it must be clear that $x = 2$ is the only answer If x is eliminated: M1 elimination M1 obtain quadratic in y M1 solve for y A1 complete to a single value of x	

Type MS · Sketching a rational curve with its asymptotes

Answer 1 of 17 · Question 9 part b · 5 marks

2020 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dV}{}$	

Type MS · Sketching a rational curve with its asymptotes

Answer 2 of 17 · Question 3 part d · 4 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

3 a	$\frac{a}{-2}=4 \Rightarrow a=-8$	M1 A1 (2)
b	$x=\frac{1}{2}$	B1 (1)
c (i)	$\left(\frac{3}{-"-8"},0\right)\left(=\left(\frac{3}{8},0\right)\right)$	B1 ft
c (ii)	$(0,3)$	B1 (2)
d	Asymptote drawn at $y=4$. The asymptote must either be clearly labelled with its equation or pass through the y axis with 4 clearly labelled.	B1
	Asymptote with equation drawn at $x=\frac{1}{2}$ The asymptote must either be clearly labelled with its equation or pass through the x axis with $\frac{1}{2}$ labelled.	B1 ft
	Correct curve drawn with two branches	B1 ft
	"$\frac{3}{8}$" and "3" labelled on x and y axes respectively	B1 ft (4)
		Total 9 marks

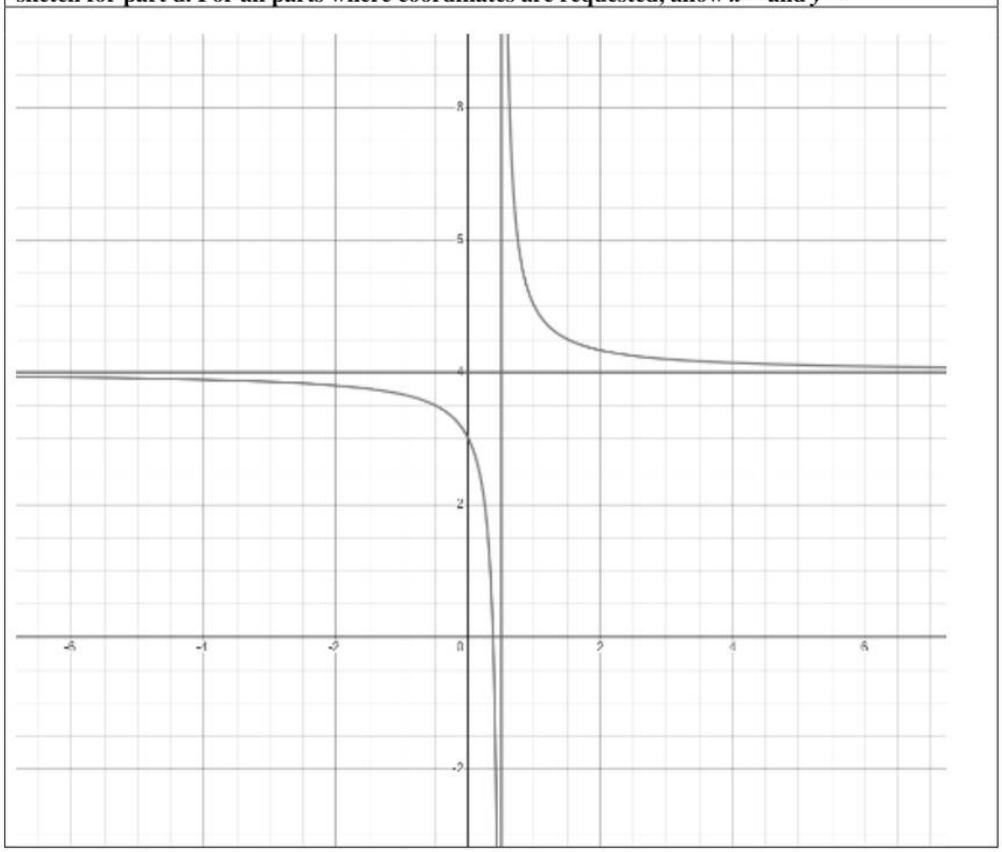
Part	Mark	Additional Guidance
(a)	**M1**	For $\frac{a}{-2}=4$
	A1	For $a=-8$
	Note: it is acceptable if a candidate uses a suitably large value for x and deduces the correct value for a. M1 A1	
(b)	**B1**	For $x=\frac{1}{2}$
(c)(i)	**B1 ft**	For $\left(\frac{3}{-"-8"},0\right)\left(=\left(\frac{3}{8},0\right)\right)$ ft their value from (a). Allow omission of brackets. This mark is awarded for the point at which you see $\left(\frac{3}{-\text{their } a},0\right)$ May be written $x=$
(ii)	**B1**	For $(0,3)$. Allow omission of brackets. May be written $y=$
If there is no labelling of (i) and (ii), it must be unambiguously indicated which coordinate crosses the y axis and which crosses the x axis. If any doubt, do not award the marks.		
(d)	**B1**	For asymptote with equation drawn at $y=4$. The line must be labelled with the equation or must clearly pass through the y-axis labelled as 4. There must be at least one branch of the curve present, which must not cross or bend back from the asymptote.
	B1 ft	For asymptote with equation drawn at $x=\frac{1}{2}$, ft their answer from part b The line must be labelled with the equation or must clearly pass through the x-axis labelled as 0.5. There must be at least one branch of the curve drawn, which must not cross or bend back from the asymptote.

Type MS · Sketching a rational curve with its asymptotes

Answer 2 of 17 · Question 3 part d · 4 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

	B1 ft	For correct curve drawn with two branches, in the correct quadrants, ft their answers from part a and b. The curve must be drawn in quadrants 1 and 3.
	B1 ft	For a curve passing through "$\frac{3}{8}$" and "3" labelled on x and y axes respectively, ft their answers from part a, b and c.
Note: answers for parts a, b and c cannot be awarded marks for correct values labelled on the sketch for part d. For all parts where coordinates are requested, allow $x =$ and $y =$		



Type MS · Sketching a rational curve with its asymptotes

Answer 2 of 17 · Question 3 part d · 4 marks

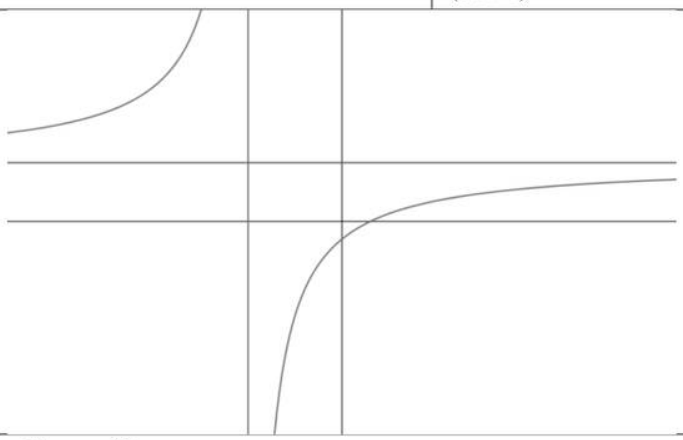
2023 JAN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 3 of 17 · Question 10 part e · 3 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 4 · find part e in this mark scheme extract

10(a) Mark (i) and (ii) together	$-\frac{1}{4}=-\frac{2}{b}\rightarrow b=8$ * $-\frac{4}{3}a+8=0$ oe or $-\frac{b}{a}=-\frac{4}{3}$ $a=6$	B1*cso M1 A1 [3]
(b)	$y=\frac{5}{"6"}$	B1ft [1]
(c)	$\left(\frac{2}{5},0\right)$	B1 [1]
(d)	$\frac{5(6x+8)-6(5x-2)}{(6x+8)^2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion ALT using product rule $(5x-2)(6x+8)^{-1}$ $5(6x+8)^{-1}+(-1)(6)(5x-2)(6x+8)^{-2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion	M1 dM1 A1 A1 [4]
(e)		B1 (curve) B1ft (asymp-totes) B1ft (intersect-ions with axes [3]
(f)	$\frac{13}{100}=\frac{52}{(6x+8)^2}$ oe $(6x+8)^2=400$ **$3x^2+8x-28=0$ oe** $x=2, y=\frac{2}{5}$ $x=-\frac{14}{3}, y=\frac{19}{15}$ oe $\sqrt{\left(("2")-\left("-\frac{14}{3}"\right)\right)^2+\left(\left("\frac{2}{5}"\right)-\left("\frac{19}{15}"\right)\right)^2}$ 6.72	M1 dM1 A1 A1 dM1 A1 [6]
		Total 18 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 3 of 17 · Question 10 part e · 3 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 4 · find part e in this mark scheme extract

Part	Mark	Additional Guidance
(a) (i) and (ii)	B1*cso	For a minimum of the steps shown in the MS to find b, no errors.
	M1	For correct substitution of $x=-\frac{4}{3}$ and $b=8$ into $ax+b=0$
	A1	Correct a.
		Students can complete this in a different order $-\frac{4}{3}a+8=0 \Rightarrow a=6$ $-\frac{b}{a}=-\frac{4}{3}\left(\text{or } -\frac{b}{6}=-\frac{4}{3}\right)\Rightarrow b=8$ B1 M1 A1
(b)	B1ft	Follow through their value of a. $y=\frac{5}{"their 6"}$
(c)	B1	$\left(\frac{2}{5},0\right)$ allow $x=\frac{2}{5}$ $y=0$ $x=\frac{2}{5}$ alone is insufficient
(d)	M1	Attempt the quotient rule. Numerator must be the difference of two terms of the form $A\times(ax+8)-B\times(5x-2)$ or $B\times(5x-2)-A\times(ax+8)$, A and $B>1$. Denominator must be of the form $(ax+8)^2$ $a\neq 0$ Allow follow through of their value of a (general principle of method mark).
	dM1	Either term on the numerator fully correct (either way round), dependent on previous method mark, doesn't need to be simplified. Denominator must be of the form $(ax+8)^2$ Allow follow through of their value of a (general principle of method mark).
	A1	$\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ eg $\frac{13}{(3x+4)^2}$ A simplified expression (denominator expanded acceptable) Dependent on both previous method marks – question states must use calculus, so some method must be shown.
	A1	$\frac{52}{(6x+8)^2}$ and correct conclusion, for example, (the numerator is a positive number) eg the denominator is always positive eg the $(ax+8)^2>0$ and therefore (the fraction) is always positive. No incorrect work for this mark eg don't accept incorrectly simplified gradient.
ALT	M1	Attempt the product rule. $C\times(ax+8)^{-1}+(5x-2)(-(1))(D)\ (ax+8)^{-2}$, C and $D>1$. Allow follow through of their value of a (general principle of method mark).
	M1	Either term correct (either way round), dependent on previous method mark, doesn't need to be simplified
	A1 A1	As main scheme

Type MS · Sketching a rational curve with its asymptotes

Answer 3 of 17 · Question 10 part e · 3 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 4 · find part e in this mark scheme extract

(e)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes $x = -\frac{8}{\text{their } a}$, $y = \frac{5}{\text{their } a}$, ft their value for a, labelled as equations or clearly passing through the correctly labelled numbers on the axis. There must be one section of a negative reciprocal curve present, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through $\left(0, -\frac{1}{4}\right)$ and $\left(\frac{2}{5}, 0\right)$ marked clearly on the graph as coordinates or crossing points on the axis. Ignore any other branch/curve/line present. Allow follow through only of their $(c, 0)$ $c \neq 0$ from part (c)
(f)	M1	Sets their differentiated function from part (d) = $\frac{13}{100}$ Allow any function = $\frac{13}{100}$ coming from working in part (d) or re-started work to find the derivative. The function may not be the original function.
	dM1	For correctly rearranging an equation of the form $\frac{13}{100} = \frac{Q}{(cx+d)^2}$ to $(cx+d)^2 = R$ using their a, or to the form $px^2 + qx + r = 0$ $p, q, r \neq 0$ with 2 out of 3 of p, q, r correct. leading to $x =$
	A1	For $x = 2, y = \frac{2}{5}$ or $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	A1	For $x = 2, y = \frac{2}{5}$ and $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	dM1	For correct use of the formula to find the length of a line, using their values for x and y. Dependent on **first** method mark only.
	A1	For awrt 6.72

Type MS · Sketching a rational curve with its asymptotes

Answer 3 of 17 · Question 10 part e · 3 marks

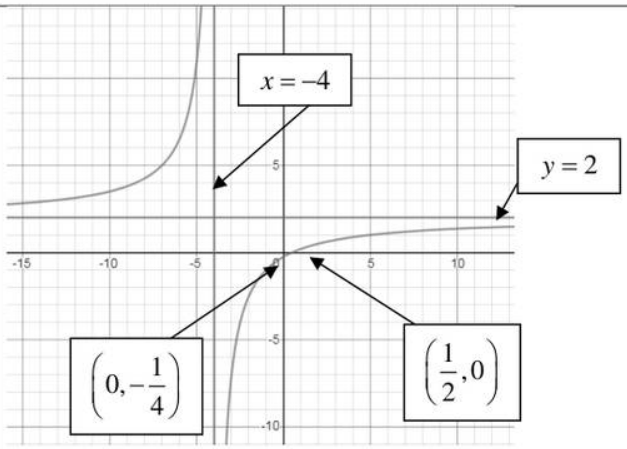
2025 JUN · Paper 02 (regional R) · extract 4 of 4 · find part e in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 4 of 17 · Question 10 part c · 3 marks

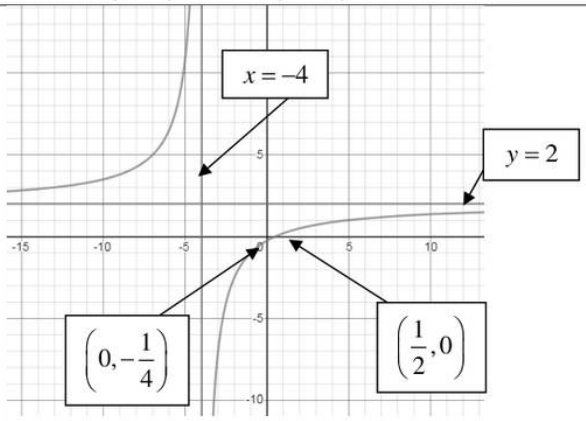
2022 JAN · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

Question	Scheme	Mark
10(a)	(i) $y=2$ (ii) $x=-4$	B1B1 [2]
(b)	$\left(\frac{1}{2},\ 0\right)$, $\left(0,-\frac{1}{4}\right)$	B1B1 [2]
(c)		B3 [3]
(d)	$\frac{dy}{dx}=\frac{2(x+4)-(2x-1)}{(x+4)^2}$ $m=1$ $\frac{dy}{dx}=1\Rightarrow\frac{9}{(x+4)^2}=1\Rightarrow(x+4)^2=9\Rightarrow x=-7,-1$ $P\ (-1,-1),\quad Q\ (-7,5)$	M1A1 B1 dM1ddM1A1 A1A1 [8]
(e)	$y=x+k$ $'-1'='-1'+k_1\Rightarrow k_1=0\quad '5'='-7'+k_2\Rightarrow k_2=12$	M1A1A1 [3]
	Total	**18 marks**

Type MS · Sketching a rational curve with its asymptotes

Answer 4 of 17 · Question 10 part c · 3 marks

2022 JAN · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1 B1	For correct **equations** only (i) $y = 2$ (ii) $x = -4$ These must be clearly labelled.
(b)	B1	For either $\left(\frac{1}{2}, 0\right)$ OR $\left(0, -\frac{1}{4}\right)$ accept $y = -\frac{1}{4}$ OR $x = \frac{1}{2}$
	B1	For both $\left(\frac{1}{2}, 0\right)$ AND $\left(0, -\frac{1}{4}\right)$ accept $y = -\frac{1}{4}$ AND $x = \frac{1}{2}$
(c)		
	B1	For the correct shape in the correct 'quadrants'. Do not allow the curves to turn back on themselves, but be reasonable in your judgement. If they all turn back on themselves clearly – withold the mark. If there is doubt over just one end, allow the mark. If you are really not sure, then please send to Review.
	B1ft	Their asymptotes drawn and labelled and there must be at least one part of the curve present that is asymptotic in nature. Any curve that crosses the asynptotes does not score this mark. Accept -4 written on the x-axis and 2 written on the y-axis.
	B1ft	Their intersections labelled. Accept $y = -\frac{1}{4}$ AND $x = \frac{1}{2}$ labelled correctly and their curve must pass through these points.
(d)	M1	For attempting to use Quotient rule: An attempt is defined as both $(2x-1)$ and $(x+4)$ differentiated CORRECTLY and the correct formula used (subtracted either way around in the numerator) with the denominator squared. $\frac{dy}{dx} = \frac{2(x+4)-(2x-1)}{(x+4)^2}$ [Correct] or $\frac{dy}{dx} = \frac{(2x-1)-2(x+4)}{(x+4)^2}$ [Incorrect] Or for an attempt to use product rule: Both terms differentiated correctly with the correct formula used. Allow a maximum of one sign error.

Type MS · Sketching a rational curve with its asymptotes

Answer 4 of 17 · Question 10 part c · 3 marks

2022 JAN · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

		$\frac{dy}{dx}=(2x-1)(1)(-1)(x+4)^{-2}+(2)(x+4)^{-1} \Rightarrow \left[\frac{dy}{dx}=\frac{-(2x-1)+2(x+4)}{(x+4)^2}\right]$
	A1	$\frac{dy}{dx}=\frac{2(x+4)-(2x-1)}{(x+4)^2}$ Fully correct
	B1	For $m = 1$
		ALT 1 for next **two marks** only
	dM1	For setting their $\frac{dy}{dx}=1$ and rearranging to reach a 3TQ. If there is no squared term in their $\frac{dy}{dx}$ then this is M0. $x^2+8x+7=0$
	ddM1	For attempting to solve their 3TQ $x^2+8x+7=0 \Rightarrow (x+1)(x+7)=0 \Rightarrow x=...,...$
		ALT 2 for next **two marks** only
	dM1	For setting their $\frac{dy}{dx}=1 \Rightarrow \frac{9}{(x+4)^2}=1 \Rightarrow (x+4)^2=9$
	ddM1	$(x+4)^2=9 \Rightarrow x+4=\pm 3 \Rightarrow x=..., ...$
	A1	For **BOTH** correct values of x, -7 and -1
	A1	For using their values of x to find either the **coordinates** of P or Q, they must be given as coordinates but allow missing or incorrect labels i.e, if they label P as Q or vice versa, allow the marks. At P $y=\frac{2\times('-1')-1}{-1+4}=-1 \Rightarrow$ Coordinates are $(-1,-1)$ **OR** At Q $y=\frac{2\times('-7')-1}{-7+4}=5 \Rightarrow$ Coordinates are $(-7,5)$
	A1	At P $y=\frac{2\times('-1')-1}{-1+4}=-1 \Rightarrow$ Coordinates are $(-1,-1)$ **BOTH** At Q $y=\frac{2\times('-7')-1}{-7+4}=5 \Rightarrow$ Coordinates are $(-7,5)$
(e)	M1	For substituting either of **their** coordinates into $y=x+k$ $'-1'='-1'+k_1 \Rightarrow k_1=...$ $'5'='-7'+k_2 \Rightarrow k_2=...$
	A1	$k_1=0$
	A1	$k_2=12$

Type MS · Sketching a rational curve with its asymptotes

Answer 4 of 17 · Question 10 part c · 3 marks

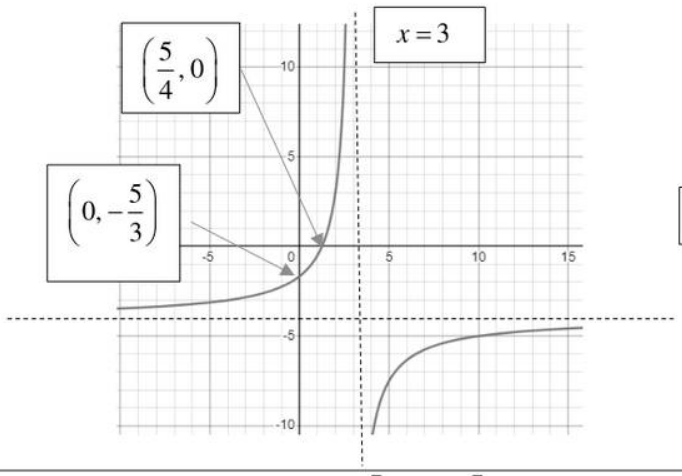
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Type MS · Sketching a rational curve with its asymptotes

Answer 5 of 17 · Question 10 part b · 5 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

10(a)	$\left(ax-5=0,\ x=\frac{5}{4}\Rightarrow\right)a=4$ $\left(\text{Line parallel to the } y\text{-axis is } x=3\Rightarrow\right)b=3$	B1 B1 {2}
(b)		B1 B1ftB1 B1ftB1ft [5]
(c)	$\left(\frac{dy}{dx}=\right)\frac{(3-x)(4)-(4x-5)(-1)}{(3-x)^2}=\left[\frac{7}{(3-x)^2}\right]$ (Gradient of l =) $\frac{7}{4}$ $\frac{7}{4}=\frac{7}{(3-x)^2}\Rightarrow(3-x)^2=4\Rightarrow 3-x=\pm 2\Rightarrow x=1,5$ When $x=1$, $y=-\frac{1}{2}$ and when $x=5$, $y=-\frac{15}{2}$ Equation of line when $x=1$: $y--\frac{1}{2}=\frac{7}{4}(x-1)\Rightarrow 4y-7x=-9$ Equation of line when $x=5$: $y--\frac{15}{2}=\frac{7}{4}(x-5)\Rightarrow 4y-7x=-65$ $\Rightarrow -65<k<-9$	M1 B1 M1A1 M1 A1 (B1B1 on ePen) ddM1 ddM1 A1 [9]

Type MS · Sketching a rational curve with its asymptotes

Answer 5 of 17 · Question 10 part b · 5 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

ALT	$(y=)\dfrac{7x+k}{4}=\dfrac{"4"x-5}{"3"-x}$ oe $4\left(\dfrac{"4"x-5}{"3"-x}\right)-7x=k$ oe	M1 A1(B1 on ePen)
	$16x-20=(7x+k)("3"-x)$ oe	
	$16x-20=21x-7x^2+3k-kx \Rightarrow 7x^2+kx-5x-20-3k(=0)$ oe	M1 A1
	$(\Rightarrow 7x^2+(k-5)x-20-3k(=0))$ oe	
	$(k-5)^2-4(7)(-20-3k)(<0)$	M1A1 (B1 B1 on ePen)
	$k^2+74k+585(<0)$	
	$(k=)\dfrac{-74\pm\sqrt{(74)^2-4\times(1)\times585}}{2}$ or $(k-9)(k-65)(=0)$	ddM1 ddM1
	$-65<k<-9$	A1 [9]

Type MS · Sketching a rational curve with its asymptotes

Answer 5 of 17 · Question 10 part b · 5 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the value of $a = 4$ or $b = 3$
	B1	For the value of $a = 4$ and $b = 3$
Note: These are independent of method marks, not given answers. Unless the answers come from **very obviously** incorrect working, these marks should be awarded for sight of the correct values.		
(b)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn and **must not** **obviously** 'bend back' on themselves. Mark intention.
	For the following marks, where candidates have used incorrect values of a and b, examiners will find using desmos.com or similar packages useful. Type into desmos $y = (ax - b)/(b - x)$ and add 'sliders' for a and b	
	B1ft	For the horizontal asymptote of $y = -4$ drawn in the correct place and labelled with its equation or where the line passes through the axis. There **must** be at least one branch of a reciprocal curve present which **must** not cross or **obviously** bend back from the asymptote(s) ignore any other curves present. Allow follow through **only** of $y = -$ (their a) and only if other conditions met.
	B1	For the vertical asymptote $x = 3$ drawn in the correct place. There **must** be at least one branch of a reciprocal curve present which **must** not cross or **obviously** bend back from the asymptote(s), ignore any other curves present.
	B1ft	A single curve (or even line) passing through $\left(0, -\frac{5}{3}\right)$ marked clearly on the graph as a coordinate or as a crossing point on the axis. Ignore any other branches present. Allow follow through only of $\left(0, -\frac{5}{\text{their } b}\right)$
	B1ft	A single curve (or even line) passing through both $\left(\frac{5}{4}, 0\right)$ and $\left(0, -\frac{5}{3}\right)$ both marked clearly on the graph as a coordinate or as crossing points on the axes. Ignore any other branches present. Allow follow through only of $\left(0, -\frac{5}{\text{their } b}\right)$

Type MS · Sketching a rational curve with its asymptotes

Answer 5 of 17 · Question 10 part b · 5 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

(c)	M1	For applying the quotient (or product rule). • With two terms in the numerator subtracted either way around. • The differentiation on both terms must be correct. • The denominator must be squared. $\frac{(3-x)(4)-(4x-5)(-1)}{(3-x)^2}$ or $\frac{(4x-5)(-1)-(3-x)(4)}{(3-x)^2}$ (if using quotient rule) Or $4(3-x)^{-1} \pm (4x-5)(3-x)^{-2}$ (if using product rule)
	B1	For (Gradient of l =) $\frac{7}{4}$
	M1	For correctly setting their gradient of l = to their derivative and an attempt to solve to find 2 values of x. Allow one error in rearrangement. Their derivative must be of the form $\frac{rx+s}{(3-x)^2}$ s **or** r could be 0, but not both
	A1	For $x = 1$ **and** 5
	M1 (B1 on ePen)	For substitution of either of their x values into the equation of their curve.
	A1 (B1 on ePen)	For $y = -\frac{1}{2}$ **and** $y = -\frac{15}{2}$
	ddM1	For correctly forming an equation of the line for either pair of their values and their gradient and rearranging (allow errors in rearrangement) to the form $4y-7x=k$. Dependent on 1st 2 method marks.
	ddM1	For correctly forming an equation of the line for both pairs of their values and their gradient and rearranging (allow rearrangement errors) to the form $4y-7x=k$ Dependent on 1st 2 method marks.
	A1	For the correct inequality as shown.
ALT	M1	For rearrangement of l to the form $y=\frac{7}{4}x+c \quad c \neq 0$ and placing equal to their equation of C or substitution of the equation for curve C in the equation for l. Allow $4\left(\frac{"4"x-5}{"3"-x}\right) = k \pm 7x$ Condone using an inequality sign, follow through their a and b
	A1 (B1 on ePen)	For a correct unsimplified equation. **DO NOT** condone using an inequality sign
	M1	For an attempt to rearrange their equation of the form $mx+c=\frac{4x-5}{3-x}$ $m, c \neq 0$, to allow them to reach a "3TQ", allow one error in rearrangement. Condone using an inequality sign By "3TQ", we means terms in x, x^2 and constant, all on 'one side' even if the equals sign is missing or an incorrect inequality sign is shown, terms simplified where possible.
	A1	For the correct "3TQ" shown (oe), the term in x does not have to be factorised and = 0 may be omitted. "3TQ" defined as above
	M1 (B1 on ePen)	For the correct formulation of the discriminant from their "3TQ". Inequality sign doesn't need to be shown or can be incorrect.
	A1 (B1 on ePen)	For the correct unsimplified discriminant. Inequality sign doesn't need to be shown or can be correct.
	ddM1	For a correct method to solve their quadratic equation in k, leading to a value of k. Dependent on first 2 method marks. Mark awarded for the correct **method.**
	ddM1	For a correct method to solve their quadratic equation in k, leading to two values of k. Dependent on 1st 2 method marks. Mark awarded for the correct **method.**
	A1	For the inequality shown.

Type MS · Sketching a rational curve with its asymptotes

Answer 5 of 17 · Question 10 part b · 5 marks

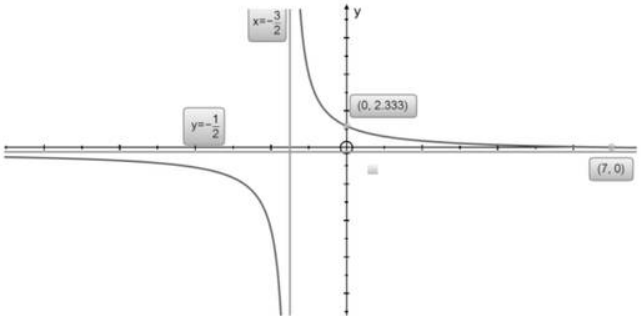
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Type MS · Sketching a rational curve with its asymptotes

Answer 6 of 17 · Question 10 part c · 3 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

10 (a)	(i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$	B1 B1 [2]
(b)	$(7,0)$ $\left(0,\frac{7}{3}\right)$	B1 B1 [2]
(c)		B1ft curve B1ft asymptotes B1ft (intersections with x- and y-axes) [3]
(d)	$\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$	M1
	$\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$	M1A1* cso [3]
	ALT – product rule	
	$-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$	M1
	$-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$	M1A1* cso

Type MS · Sketching a rational curve with its asymptotes

Answer 6 of 17 · Question 10 part c · 3 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

(e)	$-\frac{17}{(2x+3)^2}=-\frac{17}{16}$	M1
	$(2x+3)^2=16\Rightarrow 2x+3=\pm\sqrt{16}\Rightarrow x=0.5 \text{ or } -3.5$	dM1
	$y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$	ddM1
	$\left(-\frac{7}{2},-\frac{21}{8}\right)$	A1
	$y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$	M1A1
	$34x+32y+203=0$	A1 [7]
		Total 17 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 6 of 17 · Question 10 part c · 3 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.		
(a)	(i) B1	For $x=-\frac{3}{2}$ oe
	(ii) B1	For $y=-\frac{1}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given
(c)	B1ft	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(d)	M1	Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$. Denominator must be of the form $(2x+3)^2$
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants c, d.
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
(e)	M1	For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d)
	dM1	For rearranging to obtain $(2x+3)^2=16$ and attempt to solve or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve.

Type MS · Sketching a rational curve with its asymptotes

Answer 6 of 17 · Question 10 part c · 3 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract

	ddM1	For substitution of their x into the original equation to obtain y. This must **not** be $x=\frac{1}{2}$ This M mark is dependent on the **two** previous M marks
	A1	For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$
	M1	For substitution of their coordinates of P and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method. If $y=mx+c$ is used a value of c must be found **and** an equation needs to be formed. For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [
	A1	For any correct equation of the tangent with the correct values of x and y in any form.
	A1	For the correct equation of tangent given in the required form. $34x+32y+203=0$

Type MS · Sketching a rational curve with its asymptotes

Answer 6 of 17 · Question 10 part c · 3 marks

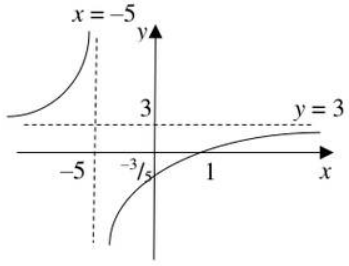
2025 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 7 of 17 · Question 3 part d · 3 marks

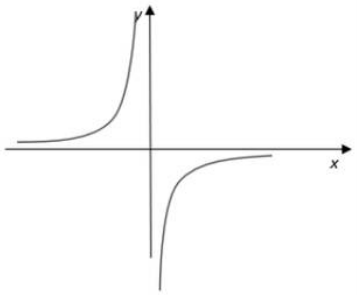
2021 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

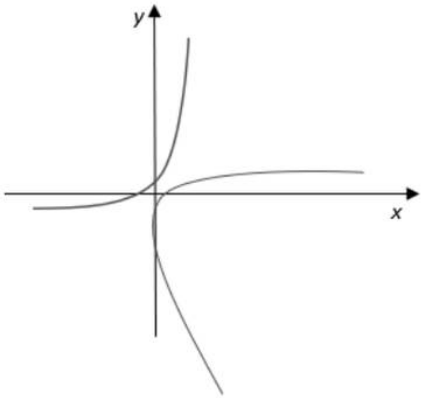
3 (a)	$\frac{dy}{dx} = \frac{a(x+5)-(ax-3)}{(x+5)^2}$	M1
	When $x = 2$ $\frac{dy}{dx} = \frac{7a-2a+3}{49} = \frac{18}{49} \Rightarrow a = ...$	M1
	$a = 3$ *	A1cso (3)
3 (b)(i)	$y = 3$	B1
(b)(ii)	$x = -5$	B1 (2)
(c)(i)	$(1, 0)$	B1
(c)(ii)	$\left(0, \frac{-3}{5}\right)$	B1 (2)
(d)	 Curve drawn Asymptotes drawn and labelled $^{-3}/_{5}$, 1 labelled on axes	B1 B1ft B1ft (3)
		[10]
		Total 10 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 7 of 17 · Question 3 part d · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>(b)(i)</td><td>B1</td><td>For $y = 3$ This must be an equation of a line. Do not award for just 3.</td></tr>
<tr><td>(ii)</td><td>B1</td><td>For $x = -5$ This must be the equation of a line. Do not award for just –5</td></tr>
<tr><td>(c)(i)</td><td>B1</td><td>For (1, 0) or clearly listing $x = 1$, $y = 0$ as a pair.</td></tr>
<tr><td>(ii)</td><td>B1</td><td>For $\left(0, -\frac{3}{5}\right)$ oe. or clearly listing $x = 0$, $y = -\frac{3}{5}$ as a pair.</td></tr>
<tr><td rowspan="4">(d)</td><td rowspan="2">B1</td><td>For the curve drawn with two branches anywhere on the grid provided it is a negative reciprocal curve. The ends of the curves must be asymptotic and must not turn back on themselves. Do not allow any obvious overlap across the ends of the curve with evidence of the presence of asymptotes.</td></tr>
<tr><td>For example, accept:
Negative reciprocal curve in incorrect position
 

Do not accept:
• Overlap of asymptotes
• Ends turning back on themselves
 
</td></tr>
<tr><td>B1ft</td><td>For their asymptotes correctly drawn and clearly labelled with their equation.
At least one branch of the curve is required. It must be a negative reciprocal and it must be in the correct position for their asymptotes. The follow through is available for their answers in part b.
If correct asymptotes appear on the sketch, do not award marks retrospective marks in part b.</td></tr>
<tr><td>B1ft</td><td>The curve must be drawn going through their two points of intersection. It must be a negative reciprocal, in the correct position for their intersections and clearly marked on the axes. The follow through is available for their answers in part c. Allow the other branch to even be missing.
If correct coordinates appear on the sketch, do not award marks retrospective marks in part c.</td></tr>
</table>

Type MS · Sketching a rational curve with its asymptotes

Answer 7 of 17 · Question 3 part d · 3 marks

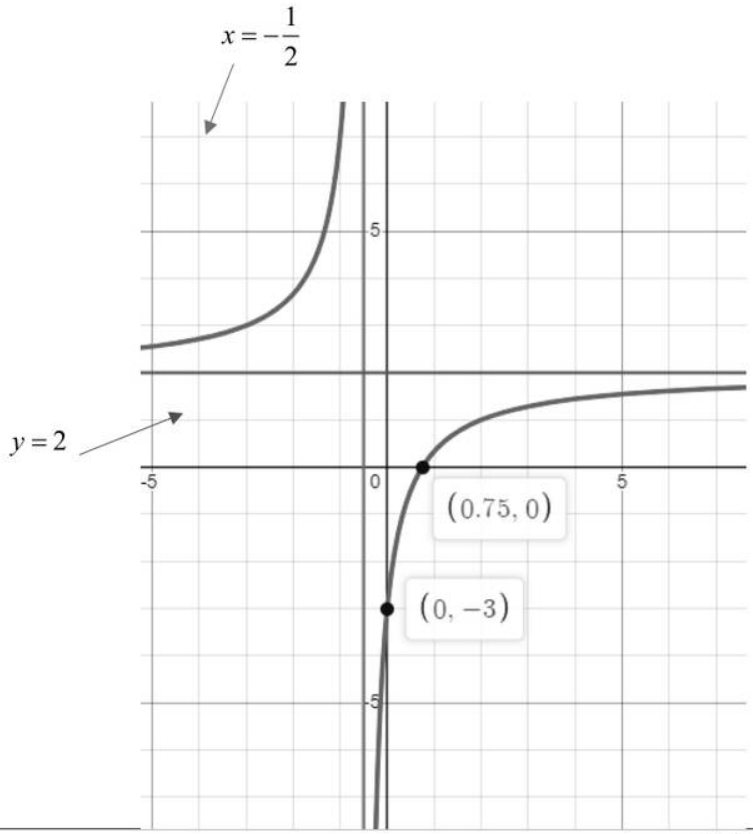
2021 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 8 of 17 · Question 3 part c · 3 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

3(a)	$\left[y = 2 - \frac{5}{2x+1} = \right] \frac{4x-3}{2x+1}$	B1
	(i) $x = -\frac{1}{2}$	B1
	(ii) $y = 2$	B1 [3]
(b)	(i) $(0, -3)$	B1ft
	(ii) $\left(\frac{3}{4}, 0\right)$ oe	B1ft [2]
(c)		B1 B1ft B1ft [3]
		Total 8 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 8 of 17 · Question 3 part c · 3 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\frac{4x-3}{2x+1}$ or $\frac{2(2x+1)}{2x+1}-\frac{5}{2x+1}$ or $\frac{4x+2-5}{2x+1}$ **Both** correct asymptote equations seen will imply this mark.
(i)	B1	For $x=-\frac{1}{2}$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(ii)	B1	For $y=2$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(b)	B1ft	For the coordinates of the y-intercept, follow through their a or b and c from any expression of the form $\frac{4x+a}{2x+1}$ $a\neq 0$ or $\frac{4x+b+c}{2x+1}$ $b,c\neq 0$ Condone eg just $y=-3$ or $y=a$ without $x=0$ shown.
	B1ft	For the coordinates of the x-intercept, follow through their a as above. ie ft $(x=)-\frac{a}{4}$ Condone $x=\frac{3}{4}$ or $x=-\frac{a}{4}$ without $y=0$ shown.
Note on labelling: Answers presented in the correct order without labelling – we would award all marks. If labelling not present, we can't assume the candidate's thinking.		
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes, ft their asymptotes listed from part (a). Labelled as equations or clearly passing through correctly labelled numbers on the axes. Must be one section of negative reciprocal curve, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied. Mark intention. Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through the correct intercepts. Ft their coordinates from part (b) marked clearly on the graph as coordinates or crossing points on the axes. Ignore any other branch/curve/line present.
	The answers to (a) and (b) cannot be retrieved from (c)	

Type MS · Sketching a rational curve with its asymptotes

Answer 8 of 17 · Question 3 part c · 3 marks

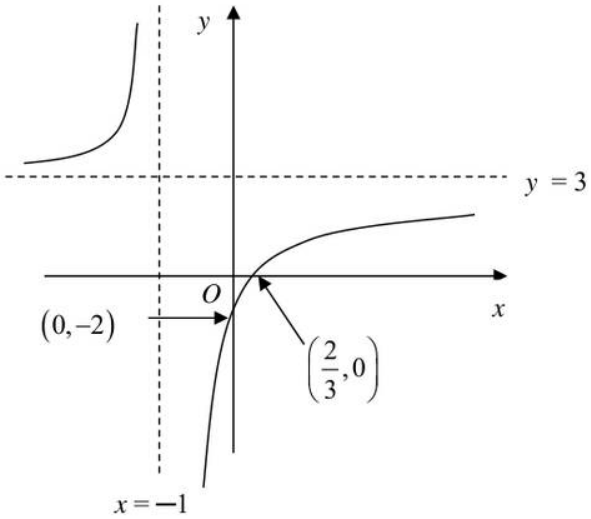
2025 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 9 of 17 · Question 11 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 5 · find part c in this mark scheme extract

11 (a)	(i) $y=3$ (ii) $x=-1$	B1B1 [2]
(b) (i) (ii)	$0=\frac{3x-2}{x+1}\Rightarrow 3x=2\Rightarrow x=\frac{2}{3}$ $y=\frac{3\times 0-2}{0+1}=-2$	B1 B1 [2]
(c)		B1 (shape and position) B1 (asymptotes) B1 (intersections) [3]
(d)	$mx+4=\frac{3x-2}{x+1}\Rightarrow mx^2+x(m+1)+6=0$ $b^2-4ac<0\Rightarrow (m+1)^2-4\times m\times 6<0\Rightarrow m^2-22m+1<0$ $(m-11)^2-120=0\Rightarrow m=11\pm\sqrt{120}$ Defines region $11-2\sqrt{30}<m<11+2\sqrt{30}$	M1 M1 M1A1 M1 depM1A1 [7]
		Total 14 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 9 of 17 · Question 11 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 5 · find part c in this mark scheme extract

(a) (i)	
B1	$y = 3$
(a) (ii)	
B1	$x = -1$
(b) (i)	
B1	$x = \frac{2}{3}$
(b) (ii)	
B1	$y = -2$
(c)	
B1	Two curves in the correct quadrant (Condone poor shape as long as intention is clear)
B1	Two asymptotes drawn and correctly labelled (Allow –1 and 3 correctly labelled on axes to count as labelled). There must be at least one part of the curve drawn
B1	Intersections labelled
(d)	
M1	Equates C and l (Condone use of $\neq$ and <)
M1	Forms a 3TQ (Condone the use of $\neq$ and <) (allow 1 error in the expansion of correct brackets or 1 error in simplifying)
M1	Use of $b^2 - 4ac < 0$ to form a 3TQ
A1	Simplify to $m^2 - 22m + 1 < 0$
M1	Solving the 3TQ If the quadratic is incorrect then the method for solving must be shown
depM1	Simplifying to the form $a \pm b\sqrt{2}$ Dependant on previous M1
A1	$11 - 2\sqrt{30} < m < 11 + 2\sqrt{30}$ (Allow $a = 11, b = 30$)

Type MS · Sketching a rational curve with its asymptotes

Answer 9 of 17 · Question 11 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 5 · find part c in this mark scheme extract

Type MS · Sketching a rational curve with its asymptotes

Answer 9 of 17 · Question 11 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 5 · find part c in this mark scheme extract

Type MS · Sketching a rational curve with its asymptotes

Answer 9 of 17 · Question 11 part c · 3 marks

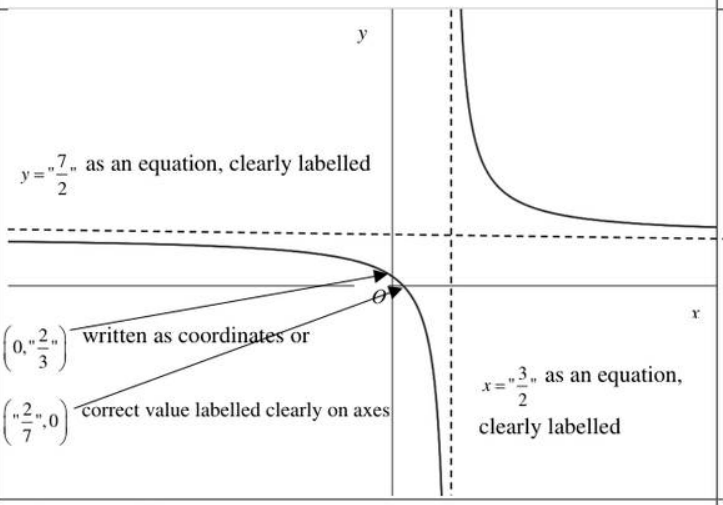
2020 JAN · Paper 02 (regional R) · extract 5 of 5 · find part c in this mark scheme extract

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Type MS · Sketching a rational curve with its asymptotes

Answer 10 of 17 · Question 10 part d · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 5 · find part d in this mark scheme extract

10 (a) (i)	$x = \frac{3}{2}$	B1
(ii)	$y = \frac{7}{2}$	B1 [2]
(b)	$\left(\frac{2}{7}, 0\right)$ $\left(0, \frac{2}{3}\right)$	B1 B1 [2]
(c)	$\frac{7(2x-3)-2(7x-2)}{(2x-3)^2}$	M1 A1
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1 [4]
	ALT – product rule	M1 A1
	$7(2x-3)^{-1} + (7x-2)(-1)(2)(2x-3)^{-2}$	
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1
(d)		B1 (curve) B1ft (asymptotes) B1ft (intersections with x- and y-axes) [3]
(e)	$-\frac{1}{17} = "\frac{-17}{(2x-3)^2}"$	M1
	"$(2x-3)^2 = 17^2$" **or** "$4x^2 - 12x - 280 = 0$" oe	dM1
	$x = 10$ $y = 4$ or $(10, 4)$	A1
	$y - "4" = 17(x - "10")$ **or** "4" = −17 × "10" + c leading to c =	M1
	$y = 17x - 166$ oe	A1

Type MS · Sketching a rational curve with its asymptotes

Answer 10 of 17 · Question 10 part d · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 5 · find part d in this mark scheme extract

	"$17x - 166" = \dfrac{7x-2}{2x-3}$ → $17x^2 - 195x + 250$	M1
	$x = \dfrac{25}{17}$ $y = -141$ or $\left(\frac{25}{17}, -141\right)$	A1 [7]
		Total 18 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 10 of 17 · Question 10 part d · 3 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 5 · find part d in this mark scheme extract

Part	Mark	Additional Guidance
	If a candidate gives **no response** to (a) and/or (b) but shows the correct answers on the graph we will award the marks. Where answers are given in (a) and/or (b) these should be marked as they stand with no reference to the graph. Ignore labelling of (i) and (ii) and mark (a) together.	
(a)(i)	B1	For $x = \frac{3}{2}$ oe
(a)(ii)	B1	For $y = \frac{7}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x = \frac{2}{7}$ and/or $y = \frac{2}{3}$ given
(c)	M1	Attempt the quotient rule. Numerator must be the difference of two terms (either way round) of the form $A.(2x-3) - B.(7x-2)$, A and $B > 1$. Denominator must be of the form $(2x-3)^2$
	A1	Either term on the numerator correct (either way round), dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x-3)^{-1} + d(7x-2)(2x-3)^{-2}$ for constants c, d.
	A1	Either term correct, dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
(d)	B1	Two branches drawn in the correct two "quadrants" created by the two aymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes. Allow BOD if intention is for curve to run alongside asymptote but there is a slight deviation back on itself.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(e)	M1	Sets their differentiated function from part (c) = $-\frac{1}{17}$
	dM1	Rearranges to get to an equation of the form shown with no denominators or a 3TQ and solves using an acceptable method to obtain $x = \ldots$. Dependent on previous method mark
	A1	Correct values for point A (10, 4)
	M1	Uses their values for x and y (from an attempt at working with gradient of the curve) with gradient 17 to find an equation for l (if using $y = mx + c$, must be a complete method arriving at $c =$) If correct $c = -166$.
	A1	Correct equation, any form

Type MS · Sketching a rational curve with its asymptotes

Answer 10 of 17 · Question 10 part d · 3 marks

2022 JUN · Paper 02 (regional R) · extract 4 of 5 · find part d in this mark scheme extract

	M1	Sets their equation for the normal equal to the curve, makes a **correct** rearrangement to remove any denominator and forms a 3TQ Note this method mark is not dependant.
	A1	Correct exact values for x and y

Type MS · Sketching a rational curve with its asymptotes

Answer 10 of 17 · Question 10 part d · 3 marks

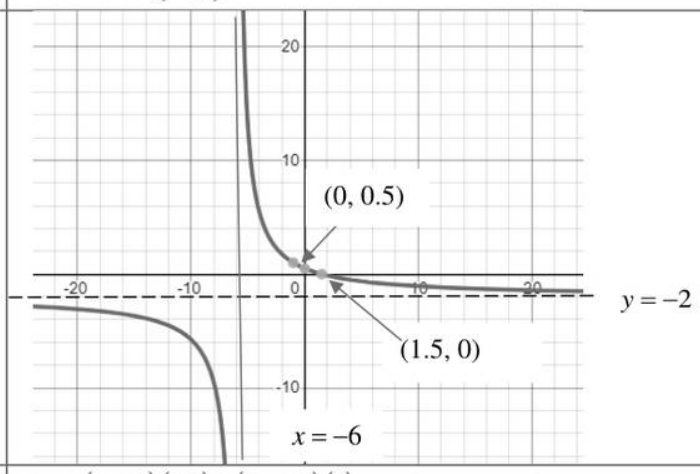
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Question number	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 11 of 17 · Question 9 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

9(a)	(i) $y=-2$ (ii) $x=-6$	B1 B1 [2]
(b)	(i) $\left(\frac{3}{2},0\right)$ (ii) $\left(0,\frac{1}{2}\right)$	B1 B1 [2]
(c)		B1 – shape B1ft– Asymptotes B1ft – Intersections [3]
(d)	$\frac{dy}{dx}=\frac{(x+6)(-2)-(3-2x)(1)}{(x+6)^2}$ $\frac{dy}{dx}=\frac{-15}{(x+6)^2}$ with conclusion; numerator negative, denominator always positive, $\frac{-}{+}\Rightarrow$ negative	M1A1 A1 [3]
(e)	$\frac{-15}{(x+6)^2}=-\frac{3}{5}\Rightarrow 25=(x+6)^2\Rightarrow x=-6\pm 5=-11,\quad -1$ $y=\frac{3-2(-1)}{-1+6}=1$ $y-1=-\frac{3}{5}(x-[-1])\Rightarrow y=-\frac{3}{5}x+\frac{2}{5}\Rightarrow k=\frac{2}{5}$	M1A1 B1 M1A1 [5]
		Total 15 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 11 of 17 · Question 9 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

Part	Mark	Notes
(a) (i) (ii)	B1	For the correct equation. If their equations are not labelled (i) and (ii) accept them in the order given only. For example, the following presentation is B0B0 $x=-6$ $y=-2$
	B1	For the correct equation.
(b)	If their coordinates are not labelled (i) and (ii) or are given in the incorrect place [for example, (b)(i) $y=\frac{1}{2}$ is B0] accept them in the order given only. For example, the following presentation is B0B0 $\left(0,\frac{1}{2}\right)$ $\left(\frac{3}{2},0\right)$	
(i)	B1	For the correct coordinates. Accept $x=\frac{3}{2}$
(ii)	B1	For the correct coordinates. Accept $y=\frac{1}{2}$
(c)	B1	For the correct shape with two branches the correct way around anywhere in the grid. It must be asymptotic in nature. The ends of the curve must not come back on themselves. Whilst you need to be fairly generous, any obvious turning back is B0
	B1ft	For the correct asymptotes drawn and labelled with at least one branch of the curve [which must be asymptotic in nature in at least one branch] in the correct place seen. Ft their asymptotes Accept the vertical line drawn shown passing through their -6 and the horizontal line drawn shown passing through their -2
	B1ft	For the correct coordinates of intersections with the relevant branch of the curve drawn correctly seen. The curve must go through the points. Do not accept touching the axis. Allow $\frac{1}{2}$ marked on y-axis and $\frac{3}{2}$ marked on the x-axis. Ft their coordinates of intersections.
(d)	M1	For an attempt at quotient rule. • The denominator must be squared, or accept $(x+6)(x+6)$ or $x^2+12x+36$ • Both $(x+6)$ and $(3-2x)$ differentiated correctly. • The two terms in the numerator subtracted either way around.
	A1	Fully correct derivative (simplification not required for this mark).
	A1	For a correct simplified derivative with a correct conclusion. For example: $(x+6)^2 \geqslant 0$, -15 is negative, $\frac{\text{negative}}{\text{positive}}$ is always negative. [Accept also $(x+6)^2>0$]

Type MS · Sketching a rational curve with its asymptotes

Answer 11 of 17 · Question 9 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

(e)	M1	For setting the value of $-\frac{3}{5}$ = their $\frac{dy}{dx}$ with an attempt to find at least one value of x. Allow this even if their derivative results in a linear equation.
	A1	For **both** correct values of x
	B1	For $y=1$ using $x=-1$ OR For $y=-5$ using $x=-11$
	M1	For forming an equation of the line with either $x=-1,\quad y=1$ or $x=-11,\quad y=-5$ or their x and their y
	A1	For the correct value of k (accept an embedded value). $y=-\frac{3}{5}x+\frac{2}{5}\Rightarrow k=\frac{2}{5}$ You can award this mark even if the previous A mark has not been scored. So, for a correct solution without showing that $x=-11$ score M1A0B1M1A1
	ALT for last 3 marks	
	B1	Sets $-\frac{3}{5}x+k=\frac{3-2x}{x+6}$
	M1	Substitutes $x=-1$ or -11 into the above equation
	A1	For $k=\frac{2}{5}$ You can award this mark even if the previous A mark has not been scored. So, for a correct solution without showing that $x=-11$ score M1A0B1M1A1

Type MS · Sketching a rational curve with its asymptotes

Answer 11 of 17 · Question 9 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 12 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

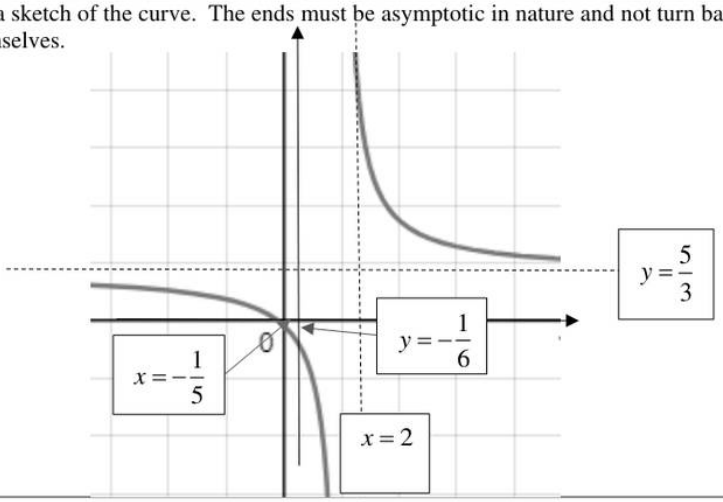
11 a	$\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$	M1A1A1 (3)
b (i)	At $x = 0$ $\frac{dy}{dx} = -\frac{11}{12}$	B1
	So $\frac{dy}{dx} = \frac{-6(2a-1)-a}{36} = -\frac{11}{12}$	M1
	$-13a+6 = -33 \Rightarrow a = 3$*	dM1 A1 cso (4)
b (ii)	$y = \frac{5x+1}{3x-6}$*	B1 cso (1)
c	Asymptote with equation drawn at $y = 2$	B1
	Asymptote with equation drawn at $x = \frac{5}{3}$	B1
	Correct curve drawn with two branches	B1
	$-\frac{1}{5}$ and $-\frac{1}{6}$ labelled on x and y axes	B1 B1 (5)
d	$\frac{12}{11}x - \frac{1}{6} = \frac{5x+1}{3x-6}$	M1
	$(3x-6)(72x-11) = 66(5x+1)$	M1
	$x(72x-265) = 0$ oe	M1
	$x = \frac{265}{72}$ oe	A1 (4)
		Total 17 marks

Part	Mark	Notes
(a)	**M1**	For attempting to use a correct quotient rule: The definition of an attempt is as follows: • The denominator must be correct • The terms in the numerator must be subtracted either way around • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $ax-6 \to \pm a$
	A1	For one term in the numerator correct
	A1	For a fully correct differentiated expression. $\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$ No simplification is required.

Type MS · Sketching a rational curve with its asymptotes

Answer 12 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

	Uses Product Rule	
	M1	For attempting to use a correct product rule: The definition of an attempt is as follows: • The terms in the numerator must be added. • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $(ax-6)^{-1} \to \pm a(ax-6)^{-2}$
	A1	For one term correct
	A1	For a fully correct differentiation $\frac{dy}{dx} = \left[(2a-1)x+1\right](a)(ax-6)^{-2} + (ax-6)^{-1}(2a-1)$
(b)(i)	**B1**	For finding the gradient of the tangent $= -\frac{11}{12}$
	M1	For using that at the point A $x = 0$ and setting their $\frac{dy}{dx} = \left(-\frac{11}{12}\right)$ which must be the gradient of their tangent to obtain $\frac{-6(2a-1)-a}{36} = '-\frac{11}{12}' \Rightarrow \left(\frac{-13a+6}{36} = -\frac{11}{12}\right)$
	dM1	For simplifying to a linear equation and attempting to solve.
	A1	For $a = 3$ with no errors seen
(b)(ii)	**B1**	For $y = \frac{5x+1}{3x-6}$ exactly in that form. You must see evidence of substitution of $a = 3$ for this mark. There must be $y =$ …. [Condone missing $x \neq 2$]
(c)		This is a sketch of the curve. The ends must be asymptotic in nature and not turn back on themselves. 
	B1	For asymptote **with equation drawn** at $y = \frac{5}{3}$ OR $x = 2$
	B1	For both asymptotes **with equations drawn** at $y = \frac{5}{3}$ AND $x = 2$
	B1	Correct curve drawn with two branches with one branch in 1st Quadrant and another branch in Quadrants 2, 3 and 4 with the graph drawn the correct way around. The graph must be asymptotic in nature.

Type MS · Sketching a rational curve with its asymptotes

Answer 12 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

	B1	$-\frac{1}{5}$ labelled on x-axis **or** $-\frac{1}{6}$ labelled on y- axis
	B1	$-\frac{1}{5}$ labelled on x axis **and** $-\frac{1}{6}$ labelled on y- axis
(d)	**M1**	For setting $\frac{12}{11}x-\frac{1}{6}=\frac{5x+1}{3x-6}$ [eliminating y from both equations]
	M1	For forming a quadratic equation, 2TQ or 3TQ in any order. $(3x-6)(72x-11)=66(5x+1) \Rightarrow 72x^2-265x=0$
	M1	For solving their QE provided it is either a 2TQ or a 3TQ
	A1	For $x=\frac{265}{72}$ oe

Type MS · Sketching a rational curve with its asymptotes

Answer 12 of 17 · Question 11 part c · 5 marks

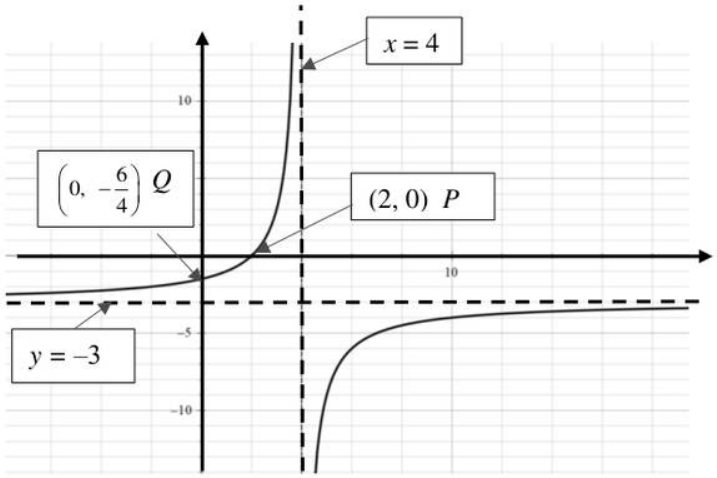
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Type MS · Sketching a rational curve with its asymptotes

Answer 13 of 17 · Question 10 part c · 3 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks
10(a) **(i)**	When $y = 0$, $x = 2$ or $(2, 0)$	B1
(ii)	When $x = 0$, $y = -\frac{6}{4}$ oe or $\left(0, -\frac{6}{4}\right)$ oe	B1 [2]
	If not labelled part (i) and (ii), do not award marks unless the candidate has presented in the correct order or has made it clear which coordinate or pair of values is *P* and which is *Q*	
(b)	(i) $x = 4$ (ii) $y = -3$	B1 B1
	If not labelled part (i) and (ii), do not award marks unless the candidate has presented in the correct order or has identification of horizontal (or parallel to *x*-axis) and vertical (or parallel to *y*-axis) is clear.	[2]
(c)		B1 B1 ft B1 ft [3]
(d)	$\left(\frac{dy}{dx} =\right) \frac{-3(x-4)-(6-3x)\times 1}{(x-4)^2} = \left[\frac{6}{(x-4)^2}\right]$ oe	M1A1A1
	where $x = 2$ $\frac{dy}{dx} = \frac{6}{(2-4)^2} = \frac{6}{4} \Rightarrow$ Gradient of normal $= -\frac{4}{6}$ oe	M1
	Equation of normal: $y - 0 = "-\frac{4}{6}"(x-2) \Rightarrow [3y = -2x + 4]$ oe	dM1A1 [6]
(e)	$\frac{6-3x}{x-4} = \frac{-2x+4}{3} \Rightarrow 2x^2 - 21x + 34 = 0$	M1
	$\Rightarrow (2x-17)(x-2) = 0$	M1
	At R $\Rightarrow x = \frac{17}{2}$, (2)	A1 [3]
		Total 16 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 13 of 17 · Question 10 part c · 3 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a) (i)	B1	Must state $y = 0$, $x = 2$ or $(2, 0)$ or clearly stating $x = 2$
(ii)	B1	Must state $x = 0$, $y = -\frac{6}{4}$ oe or $\left(0, -\frac{6}{4}\right)$ oe or clearly stating $y = -\frac{6}{4}$
(b)(i)	B1	For $x = 4$
(ii)	B1	For $y = -3$
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn and **must not** obviously 'bend back' on themselves. Mark intention.
	B1ft	For the asymptotes drawn, follow through their (b)(i) and (ii). There **must** be at least one branch of a negative reciprocal curve present in the correct place for their work in (a) and (b), which **must** not cross or obviously bend back from the asymptotes. The asymptotes **must** be labelled with their equation or shown as passing through 4 on the x-axis and -3 on the y-axis, both clearly labelled.
	B1ft	For at least one branch of a negative reciprocal curve in the correct place for their work in (a) and (b), passing through their intercepts on the axes. The intercepts should be correctly labelled with the coordinates or the axes labelled correctly with $-\frac{6}{4}$ (oe) and 2 (or their ft values) but condone labelling to be P and Q.
(d)	M1	For an expression of the form. $\frac{-a(x-4)-(6-3x)\times b}{(x-4)^2}$ oe
	A1	For an expression of the form. $\frac{-3(x-4)-(6-3x)\times b}{(x-4)^2}$ or $\frac{-a(x-4)-(6-3x)\times 1}{(x-4)^2}$ .oe
	A1	Fully correct – need not be simplified.
	M1	For substituting $x = 2$ into their $\frac{dy}{dx}$ and finding the gradient of the normal. This is not a dependent method mark, but the substitution must be into a **changed** function. If their expression does not allow substitution of $x = 2$, this mark cannot be awarded.
	dM1	For a complete and correct method to find the equation of the normal using their (changed gradient), $y = 0$ and $x = 2$. Dependent on the previous mark. If $y = mx + c$ is used they must find a value for c.
	A1	For any correct equation. **This can be in any form and may be left unsimplified.**
(e)	M1	For equating their equation of the normal to C and attempting to form a 3TQ. The attempt must involve correctly removing both denominators of the equation as a minimum and any attempt to collect terms.
	M1	For a minimally acceptable (see general guidance) and complete attempt to solve their quadratic equation, leading to a value of x..
	A1	For the x coordinate of point R $x = \frac{17}{2}$

Type MS · Sketching a rational curve with its asymptotes

Answer 13 of 17 · Question 10 part c · 3 marks

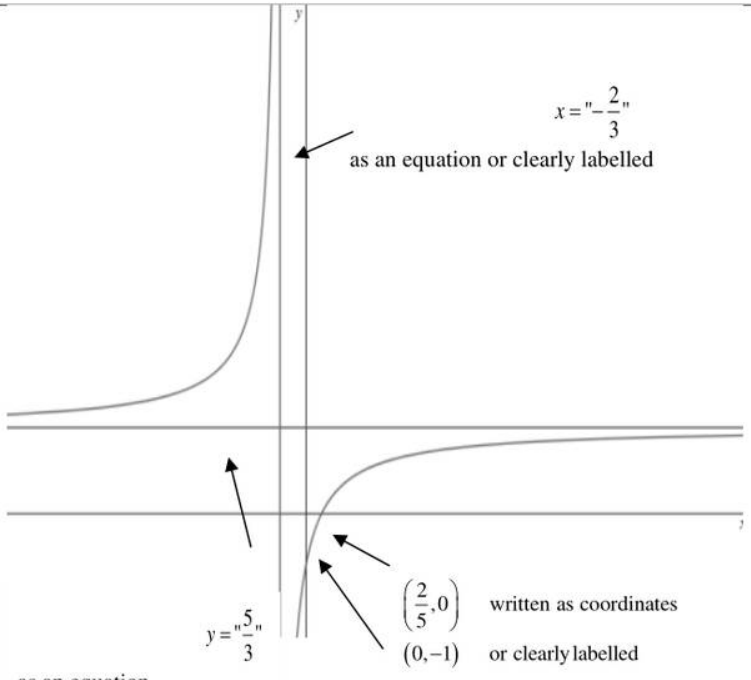
2023 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Sketching a rational curve with its asymptotes

Answer 14 of 17 · Question 10 part c · 3 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

10 (a)	(i) $\left(\frac{2}{5},0\right)$ (ii) $(0,-1)$	B1 B1 [2]
(b) (i)	$y=\frac{5}{3}$	B1
(ii)	$x=-\frac{2}{3}$	B1 [2]
(c)		B1 B1ft B1ft [3]
(d)	$\left\{y=\frac{1}{4}x+\frac{7}{4}\right\}\Rightarrow\frac{1}{4}$	B1
	$\left\{\frac{dy}{dx}=\right\}\frac{5(3x+2)-3(5x-2)}{(3x+2)^2}\left\{=\frac{16}{(3x+2)^2}\right\}$	M1 dM1A1
	$\text{"}\frac{1}{4}\text{"}=\text{"}\frac{16}{(3x+2)^2}\text{"}\Rightarrow x=...$	M1
	$x=2$ and $y\left\{=\frac{5\times\text{"}2\text{"}-2}{3\times\text{"}2\text{"}+2}\right\}=1$	A1
	$y-\text{"}1\text{"}=\text{"}-4\text{"}(x-\text{"}2\text{"})$	M1 dM1
	$(0,9)$ and $\left(\frac{9}{4},0\right)$	A1ft
	$\left\{\lvert DE\rvert\right\}=\sqrt{9^2+\left(\frac{9}{4}\right)^2}=\frac{9\sqrt{17}}{4}$	M1A1
		[11]
		Total 18 marks

Type MS · Sketching a rational curve with its asymptotes

Answer 14 of 17 · Question 10 part c · 3 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)(i)	B1	$\left(\frac{2}{5},0\right)$ Accept $x=\frac{2}{5},\{y=0\}$
(a)(ii)	B1	$(0,-1)$ Accept $\{x=0\}, y=-1$, if not labelled, we score by their order
(b)(i)	B1	Correct equation $y=\frac{5}{3}$
(b)(ii)	B1	Correct equation $x=-\frac{2}{3}$
(c)	B1	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (a), at least one section of their curve must pass through one of these intersections.
	B1ft	Two clearly marked asymptotes, ft their (b), labelled as described, there must be at least one section of the curve present, not crossing the curve.
(d)	B1	$\frac{1}{4}$ seen any where in part (d)
	M1	Attempt to differentiate y **Quotient rule**: Numerator must be the difference of two terms (either way round) of the form $P(3x+2)-Q(5x-2),\quad P,Q>0$. Denominator must be correct. **Product rule**: obtains form $P(3x+2)^{-1}-Q(5x-2)(3x+2)^{-2},\ P,Q>0$
	dM1	**Quotient rule**: Either term in the numerator correct **Product rule**: Either term correct Dependent on previous method mark.
	A1	Correct $\frac{dy}{dx}$. May be unsimplified.
	M1	Sets their $\frac{1}{4}$ equal to their $\frac{dy}{dx}$ and attempts to solve for x
	A1	$x=2$ *and* $y=1$
	M1	Finds the gradient of the normal using $\frac{-1}{their\ \frac{1}{4}}$
	dM1	A complete and correct method to find the equation of the normal. If using $y=mx+c$, c must be found. Dependent on previous method mark.
	A1ft	$(0,9)$ and $\left(\frac{9}{4},0\right)$, accept $y=9,\ x=\frac{9}{4}$ Ft their normal equation.
	M1	Correct use of the formula for the length of a line segment, using their values from their points of D and E, may not come from the normal.
	A1	Correct exact length in simplest form

Type MS · Sketching a rational curve with its asymptotes

Answer 14 of 17 · Question 10 part c · 3 marks

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Type MS · Sketching a rational curve with its asymptotes

Answer 15 of 17 · Question 9 part d.i · 0 marks

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Type MS · Sketching a rational curve with its asymptotes

Answer 16 of 17 · Question 9 part d.ii · 0 marks

2020 NOV · Paper 02 · find part d.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Sketching a rational curve with its asymptotes

Answer 17 of 17 · Question 9 part d.iii · 5 marks

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